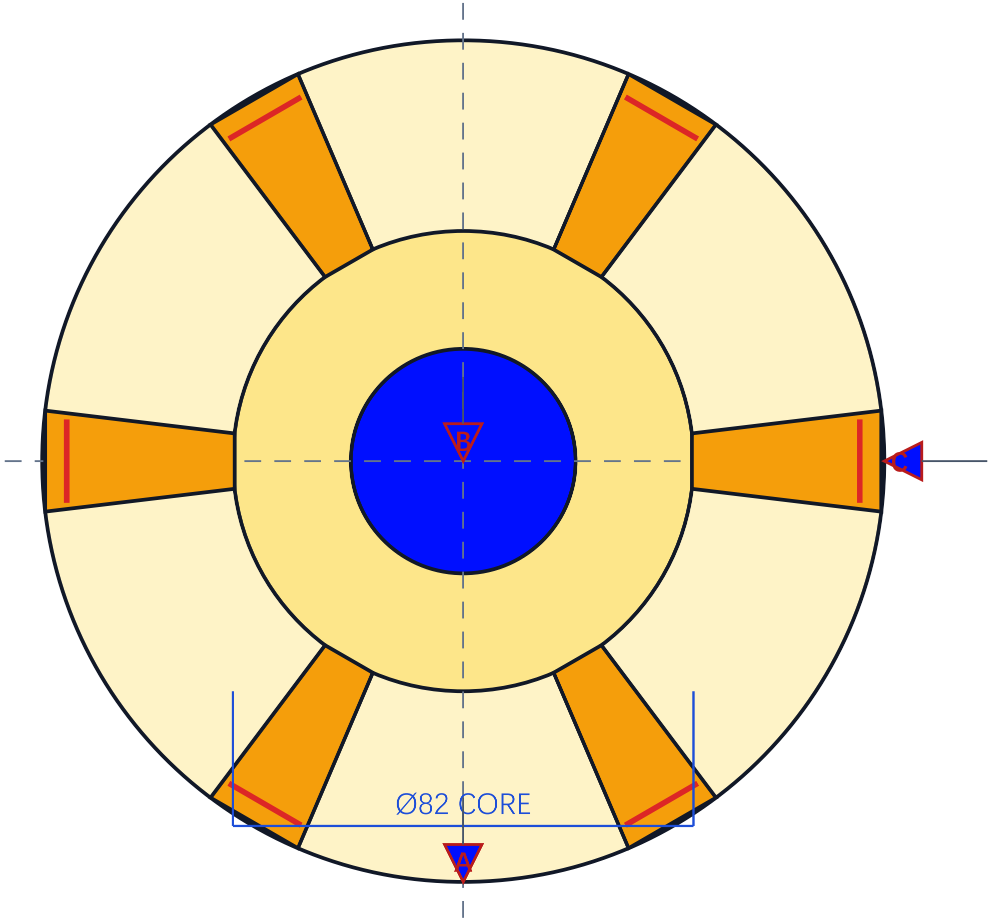
轴承座零件图 · 六翼柔顺方案

HJ-DRW-001

BEARING SEAT / PLAN VIEW / ALL DIMENSIONS IN mm



BORE Ø40

WING OUTER R74.98; LIMIT 74.97~74.99

WING WIDTH 18

RADIAL SLOT 4 WIDE

6 × 18 WELD SEGMENTS

DATUM A: shell seating plane

DATUM B: shell theoretical axis

DATUM C: independent clocking feature

CONTROLLED: Ø40 bore axis

POSITION TOLERANCE

POSITION	Ø0.05	A	B

MATERIAL: QT450-10 (actual grade to be verified by certificate)

STATUS: DESIGN-REVIEW

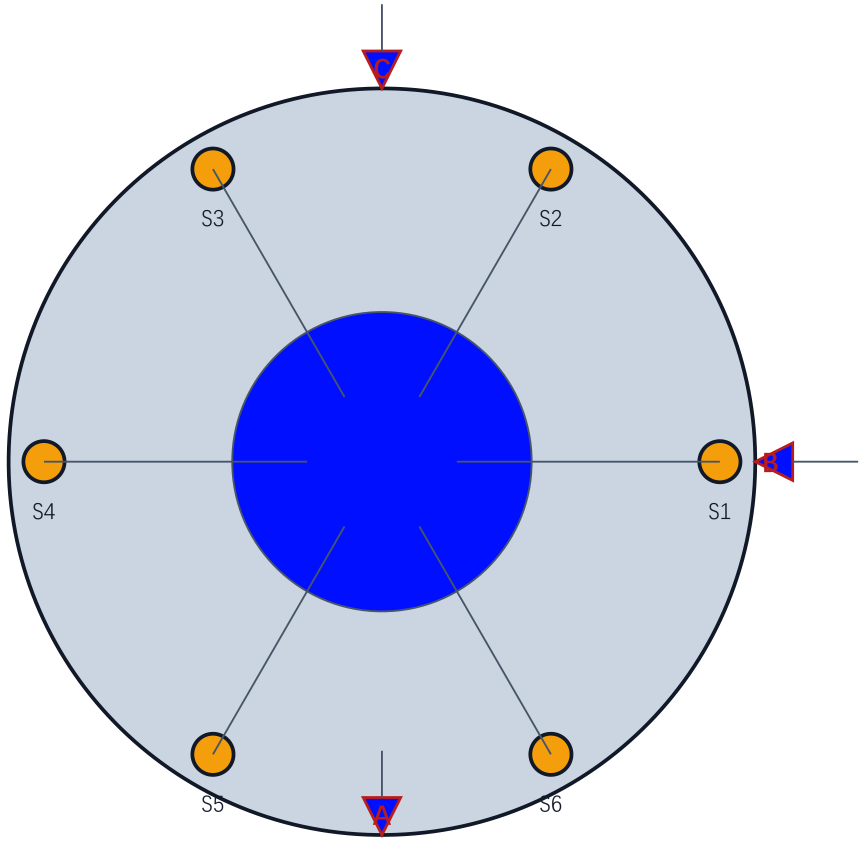
GENERAL TOLERANCE EXCLUDES FIT/CONTROLLED DIMS: ±0.10 mm (design assumption; verify before release)

Functional assembly datums: A=seat/support plane; B=shell theoretical axis; C=independent clocking; design-review only.

夹具总装图

FIXTURE ASSEMBLY / TAPERED MANDREL + SIX SUPPORTS

HJ-DRW-005



FIXTURE DEFINITION

BASE Ø190

DIAMETER TAPER (D-d)/L = 0.020 = 1:50

Ø39.90 ~ Ø40.10

Axial clamping force 500 N

Positioning repeatability 0.008 mm

6 radial compliant supports @ 1200 N/mm equiv

DOF CONTROL / ASSEMBLY MAPPING

A base top plane → assembly A

B mandrel axis → assembly B (self-centering)

C independent clocking pin → assembly C

Tapered fit auto-centers bore axis to fixture axis

MATERIAL: Fixture steel / material and heat treatment TBD

STATUS: DESIGN-REVIEW

GENERAL TOLERANCE EXCLUDES FIT/CONTROLLED DIMS: ±0.10 mm (design assumption; verify before release)

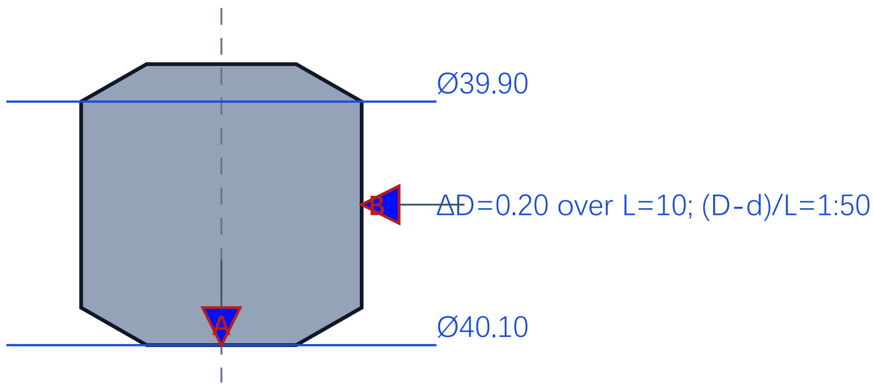
Functional assembly datums: A=seat/support plane; B=shell theoretical axis; C=independent clocking; design-review only.

夹具锥形心轴与底板零件图

HJ-DRW-006

FIXTURE PART / TAPERED MANDREL + BASE / DESIGN-REVIEW

TAPERED MANDREL - SECTION



MANDREL NOTES

Tapered fit auto-centers bore to mandrel axis

Axial clamping force: 500 N

Positioning repeatability: 0.008 mm

DATUM MAPPING

A: base top plane → assembly A

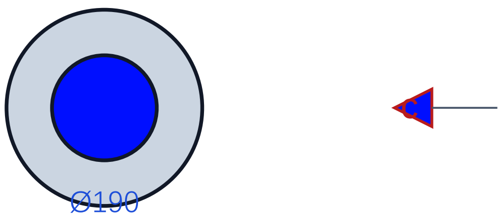
B: mandrel axis → assembly B

C: independent clocking pin → assembly C

Six support stations at 60° pitch; travel & preload TBD

No manufacturing release without tolerance stack-up

BASE PLATE - PLAN



MATERIAL: Fixture steel / material and heat treatment TBD

GENERAL TOLERANCE EXCLUDES FIT/CONTROLLED DIMS: ±0.10 mm (design assumption; verify before release)

Functional assembly datums: A=seat/support plane; B=shell theoretical axis; C=independent clocking; design-review only.

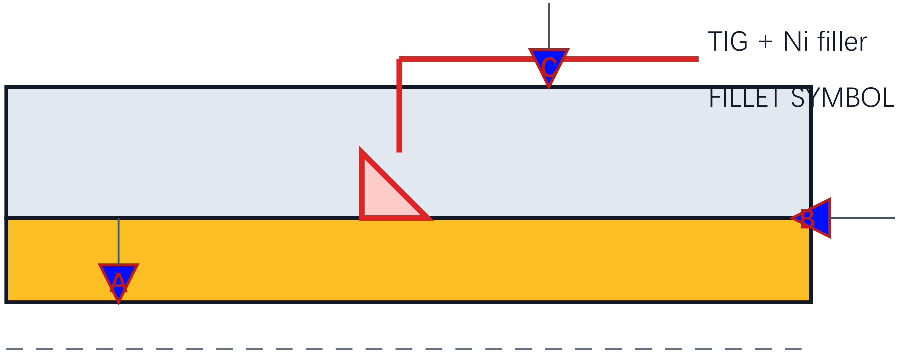
STATUS: DESIGN-REVIEW

接头与焊缝细节图

JOINT DETAIL / SECTION A-A / FILLET WELD

HJ-DRW-003

SECTION A-A



WING OD Ø149.96; LIMIT 149.94~149.98

SHELL ID Ø150.00; LIMIT 150.00~150.02

RADIAL CLEARANCE 0.01~0.04

DESIGN LEG TARGET 3.5 (UNVERIFIED)

PROCESS NOTE

Controlled limit fit; radial clearance 0.01~0.04

Fillet weld symbol per GB/T 324; TIG + Ni filler

Automatic TIG, short segments, S3 sequence

Current feed equivalent ideal leg 1.74 mm

Design/deposition geometry NOT CLOSED; macrosection required

A: shell seating plane

B: shell theoretical axis

C: independent clocking feature

Controlled Ø40 bore axis: position Ø0.05 | A | B

MATERIAL: QT450-10 + Q235B + Ni filler (weld metal grade to be verified)

GENERAL TOLERANCE EXCLUDES FIT/CONTROLLED DIMS: ±0.10 mm (design assumption; verify before release)

Functional assembly datums: A=seat/support plane; B=shell theoretical axis; C=independent clocking; design-review only.

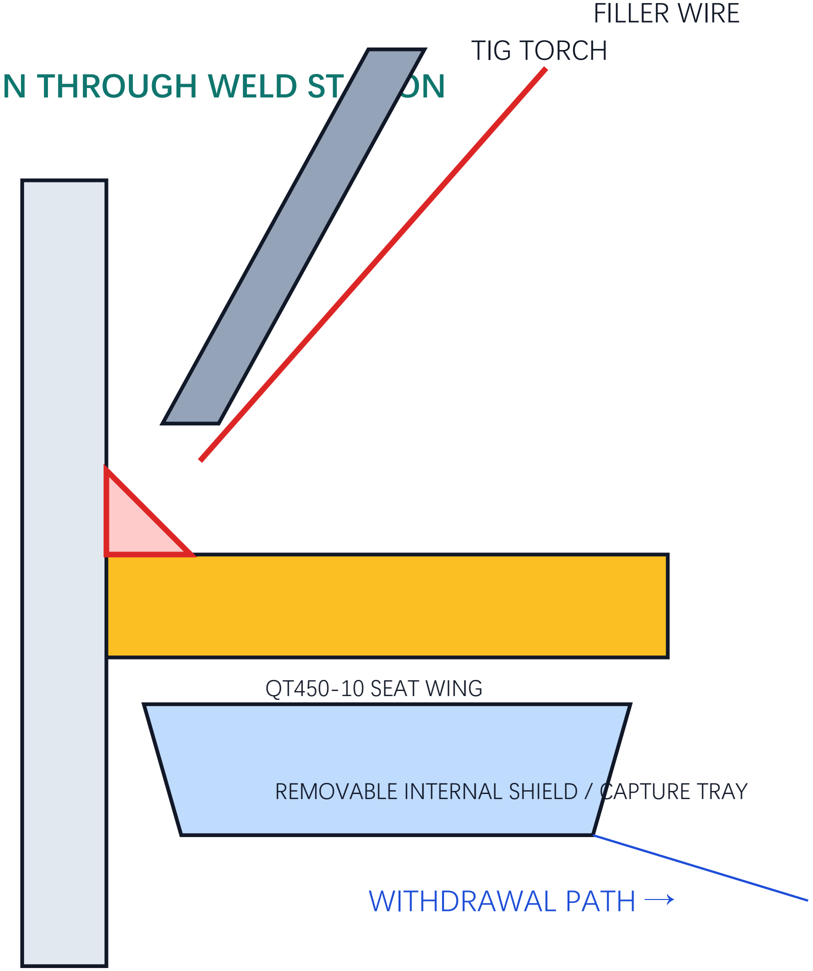
STATUS: DESIGN-REVIEW

内腔防护焊接工序总装

HJ-DRW-008

PROCESS ASSEMBLY / ACCESS + SHIELD + CONTROLLED WITHDRAWAL

SECTION THROUGH WELD STATION



CONTROLLED PROCESS

1 Insert clean, low-shedding shield before seat locking

2 Verify torch/wire/gas clearance with dry run

3 Weld with shield retained below the joint

4 Cool and unlock fixture; keep capture face upward

5 Withdraw shield along shown path without inversion

6 Inspect shield inventory and borescope internal cavity

OPEN DESIGN ITEMS

Shield material, coating, gas-flow influence and clearances TBD

No cleanliness release without real spatter/debris trial

Protect machined bore and datum surfaces during insertion

MATERIAL: Process tooling concept; material and dimensions pending access trial

GENERAL TOLERANCE EXCLUDES FIT/CONTROLLED DIMS: ±0.10 mm (design assumption; verify before release)

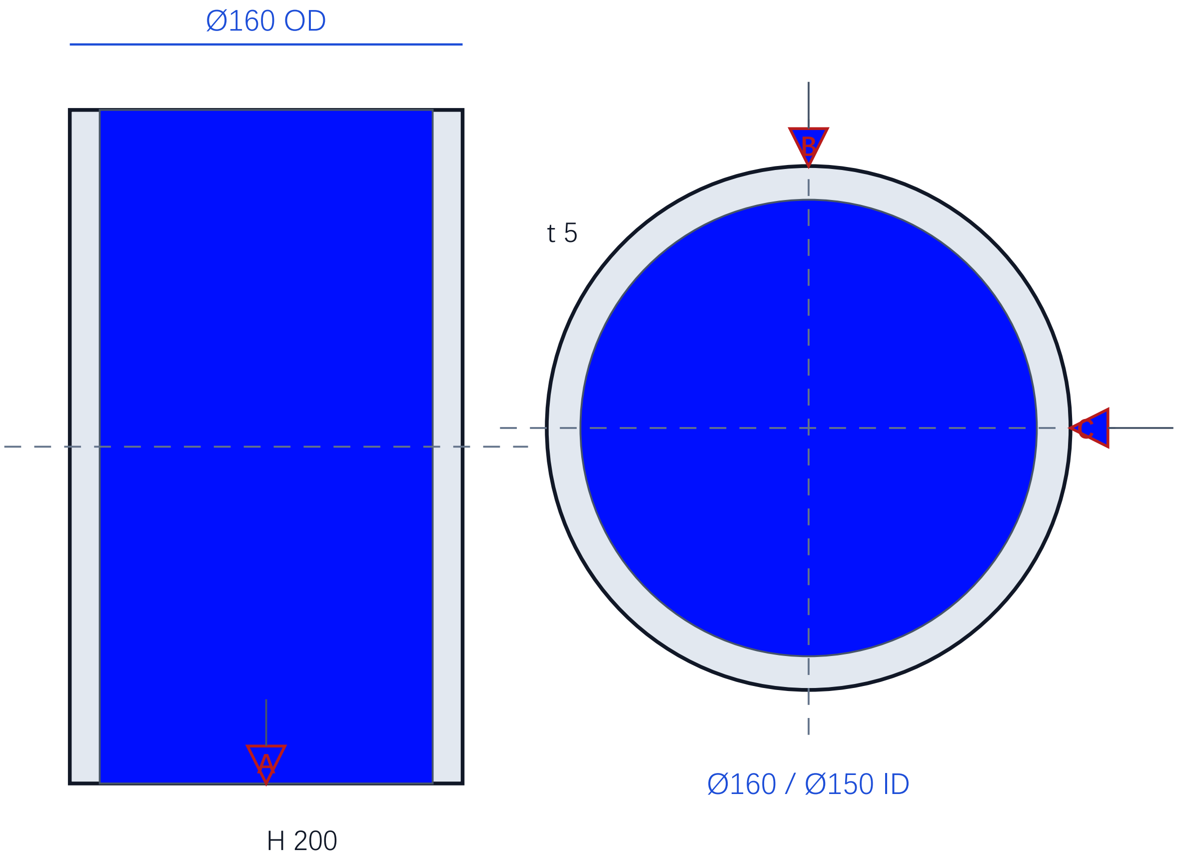
Functional assembly datums: A=seat/support plane; B=shell theoretical axis; C=independent clocking; design-review only.

STATUS: DESIGN-REVIEW

薄壁壳体零件图

HJ-DRW-002

SHELL / ORTHOGRAPHIC DEFINITION / ALL DIMENSIONS IN mm



MATERIAL: Q235B

Ø160 × H200 × t5

Cylindrical shell; seam and edge prep

DATUM A: shell seating plane

DATUM B: shell theoretical axis

DATUM C: independent clocking feature

Weld access: internal shield / borescope

MATERIAL: Q235B (actual grade to be verified by certificate)

GENERAL TOLERANCE EXCLUDES FIT/CONTROLLED DIMS: ±0.10 mm (design assumption; verify before release)

Functional assembly datums: A=seat/support plane; B=shell theoretical axis; C=independent clocking; design-review only.

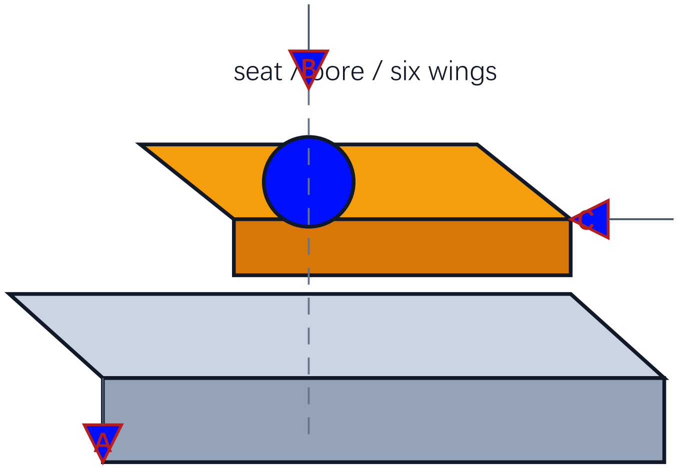
STATUS: DESIGN-REVIEW

焊接总装与定位基准图

WELD ASSEMBLY / SHELL + SEAT + FIXTURE REFERENCE

HJ-DRW-007

EXPLODED / AXONOMETRIC SCHEMATIC



fixture base + center pin

ASSEMBLY CONTROL

SHELL: Ø160 × H200 × t5

SEAT: bore Ø40; core Ø82

WELDS: 6 × 18; sequence S3

A shell seating plane / B shell theoretical axis / C independent clocking

Controlled feature: Ø40 bore axis

Inspection hand-off: visual → bore gauge/CMM → records

WPS qualification, material certificates and actual weld data remain open

Position tolerance target: Ø0.05 | A | B

POSITION	Ø0.05	A	B

MATERIAL: Q235B shell + QT450-10 seat + Ni filler + fixture steel TBD

STATUS: DESIGN-REVIEW

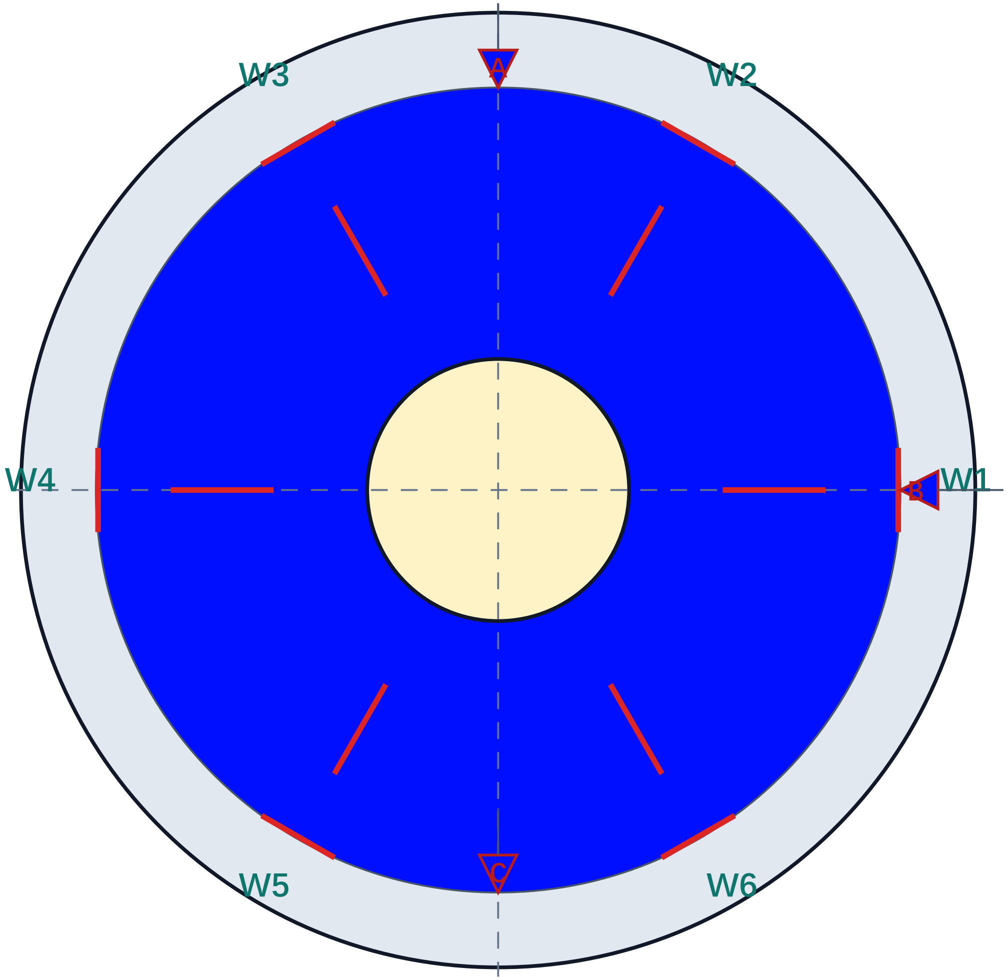
GENERAL TOLERANCE EXCLUDES FIT/CONTROLLED DIMS: ±0.10 mm (design assumption; verify before release)

Functional assembly datums: A=seat/support plane; B=shell theoretical axis; C=independent clocking; design-review only.

焊缝布置与顺序图

HJ-DRW-004

WELD LAYOUT / PLAN VIEW / S3 PATH REPRESENTATION



SEQUENCE S3

W1 → W4 → W3 → W6 → W2 → W5

6 × 18 mm short welds

Symmetric alternation; confirm start point

WELD SYMBOL: fillet / short segment

Thermal record: current, voltage, speed, interpass

A: shell seating plane

B: shell theoretical axis

C: independent clocking feature / W1

MATERIAL: Welded joint: QT450-10 / Q235B / Ni filler

STATUS: DESIGN-REVIEW

GENERAL TOLERANCE EXCLUDES FIT/CONTROLLED DIMS: ±0.10 mm (design assumption; verify before release)

Functional assembly datums: A=seat/support plane; B=shell theoretical axis; C=independent clocking; design-review only.